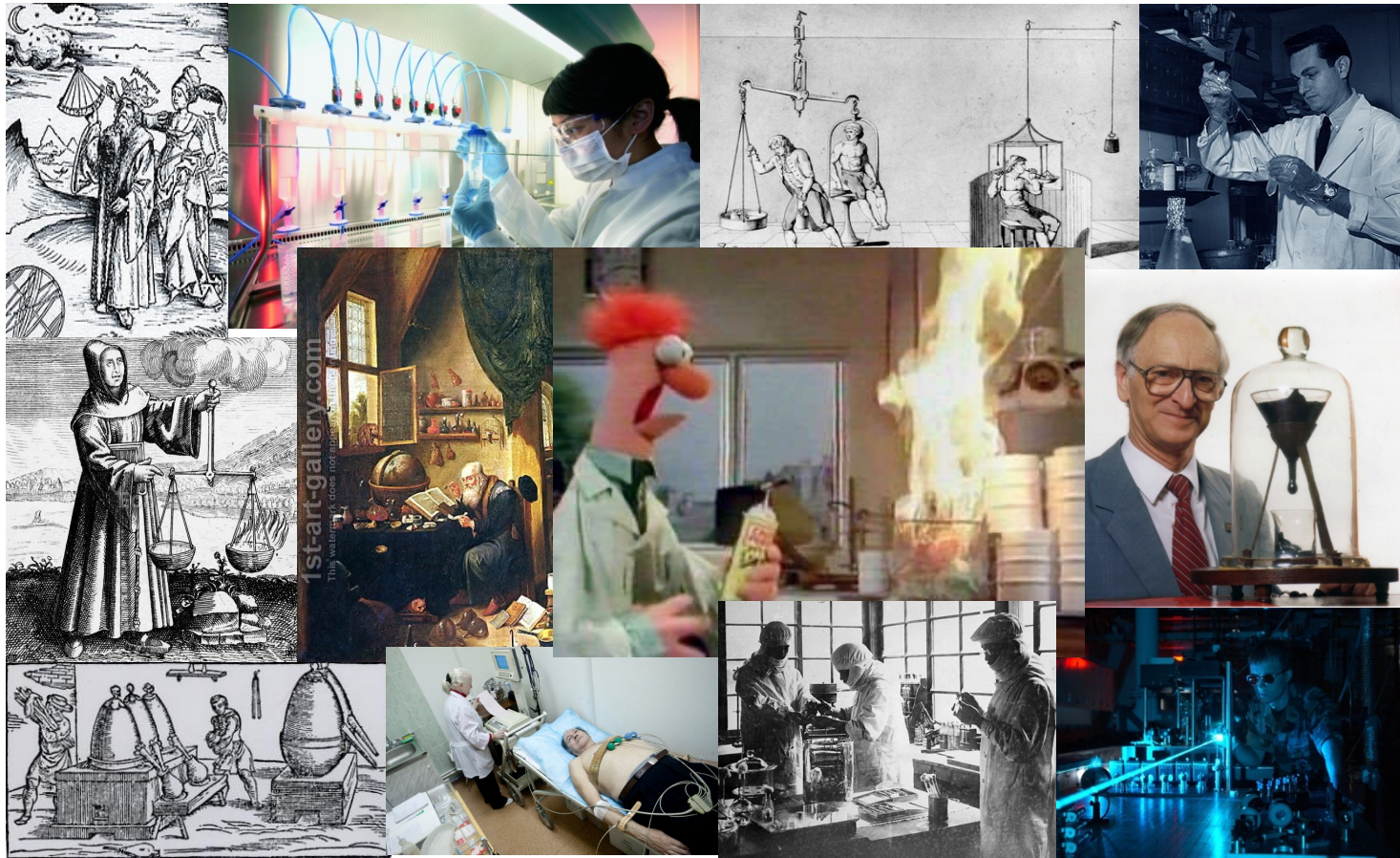


The nature of Simulation

- A possible alternative, in case the analytical solution is not feasible or too simplistic for the required complexity, is *experimentation*



The nature of Simulation

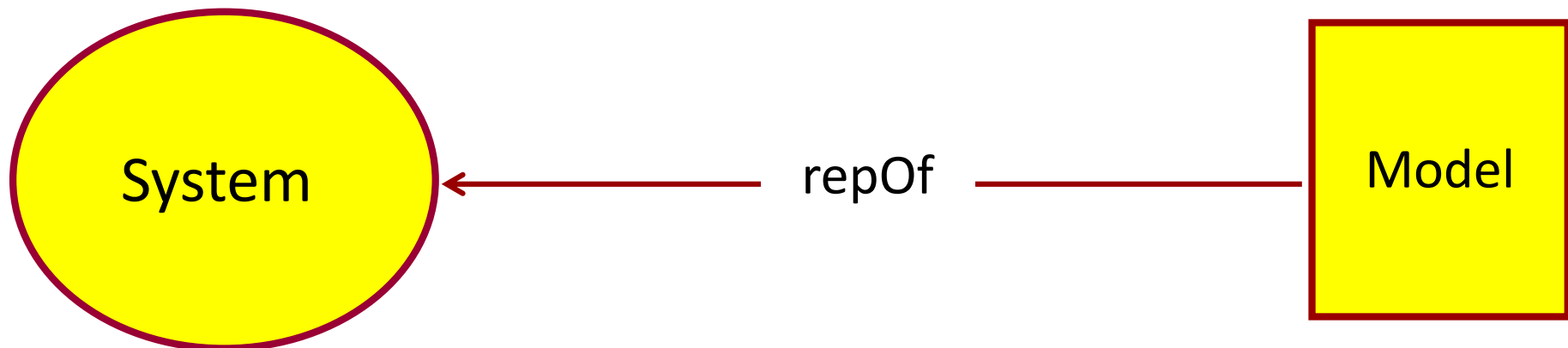
- *Simulation* is thus solicited when:
 - The mathematical model:
 - is hard or not possible to be defined and/or to be solved analytically
 - does not take into account the peculiar complexity of the problem under study
 - experimentation is:
 - not possible, because the system, process or phenomenon under study does not exist yet
 - not safe, because of the risk of social or economic losses
 - not convenient, because too expensive to be carried out
- The term in its scientific connotation dates back to 40's, when *Fermi, von Neumann e Ulam* introduced a numerical technique (later on coined as "Monte Carlo analysis") to solve nuclear problems that were too expensive to address by experimentation and too complex to address analytically

Simulation is just a part of the story..

- A **simulation** must always have a **model** and *modeling is an essential part of a simulation*
- To emphasize the modeling involved in a simulation, simulation is commonly referred to as **Modeling and Simulation (M&S)**
- The Society for Computer Simulation International (SCS) has changed its name in 2002 into *Society for Modeling & Simulation International* (www.scs.org)
- Modeling and simulation (M&S) constitutes the central part of our thinking process
 - we think by “constructing mental *models* and then *simulating* (i.e., executing) them in order to draw conclusions or make decisions”

The essential role of modeling

- Modeling is essential to human activity because every action is preceded by the construction (implicit or explicit) of a model
- What is (in short) a model?
 - a model is an abstract (i.e., simplified) representation of a system
 - and what is (in short) a system?
 - a system is a set of elements that interact

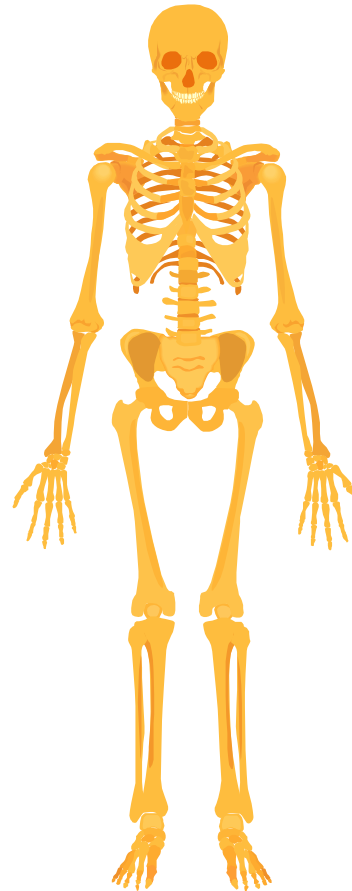


A model is a partial view of a system

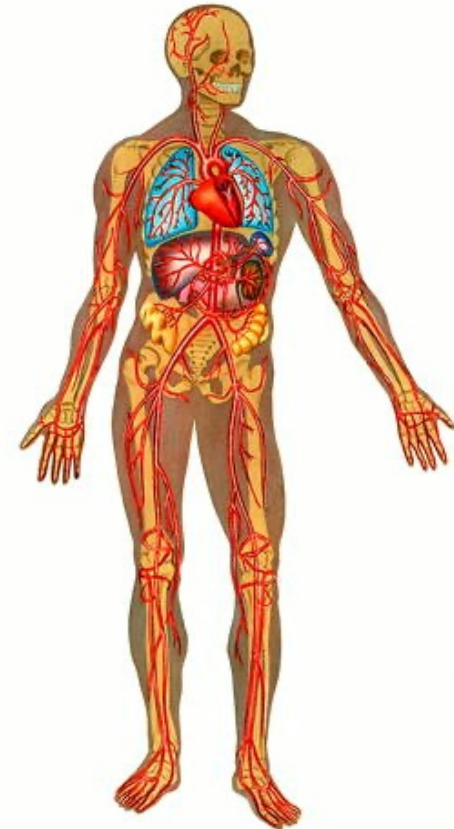
A system

represent

Several models
(partial views)



Skeleton
model



Circulatory
model

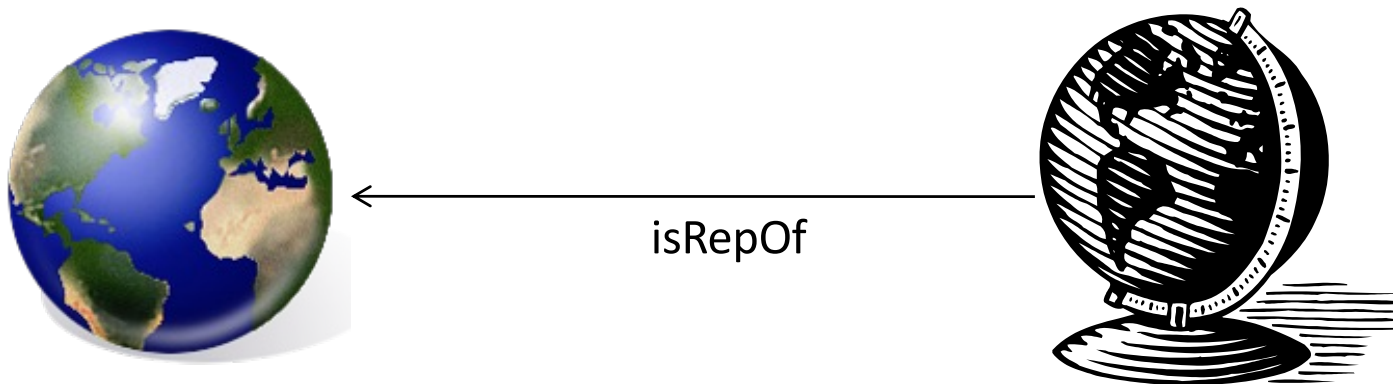
Principle of *limited substitutability*

- A model M is said to be a representation of a system S for a given set of questions Q if, for each question of this set Q, the model M will provide exactly the same answer that the system S would have provided in answering the same question



As an example...

- A globe (model of the world) can provide an answer to the question: *“may I reach Tallin from Rome without using a boat?”*
- but wouldn't be able to provide an answer to the question: *“what is the temperature in Rome?”*



The nature of Simulation

- Most complex systems require models that are also complex (to be valid)
 - Must be studied via simulation, i.e.:
 - build a model
 - implement the model
 - execute the model implementation and
 - collect data to estimate model characteristics (*simulation output analysis*)
- Example: a manufacturing company considering extending its plant
 - Build it and see if it works out?
 - Simulate current, expanded operations — could also investigate many other issues along the way, quickly and cheaply

M&S Approaches

Classification of M&S approaches:

- based on *model representation*
 - Discrete-event M&S
 - Continuous M&S
 - Monte Carlo M&S
 - Agent-based M&S
 - System Dynamics M&S
- based on *model execution*
 - Sequential simulation
 - Parallel simulation
 - Distributed/Web-based simulation
- based on *what-is-in-the-loop*
 - Hardware-in-the-loop
 - Software-in-the-loop
 - Human-in-the-loop



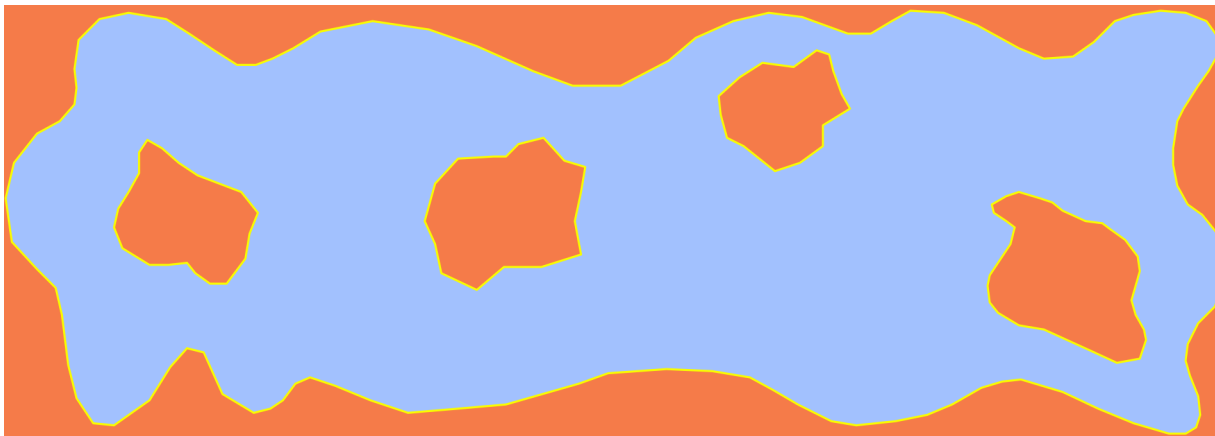
- M&S approaches differ from each other depending on how the model is *defined*, *implemented* and *executed*

Intended use

- Problem Solving
 - Training
 - Decision Making
 - Entertainment
 - Research
 - Education
- Use of single or combined M&S approaches *spans dozen of different disciplines and application domains*, for many different *intended uses*

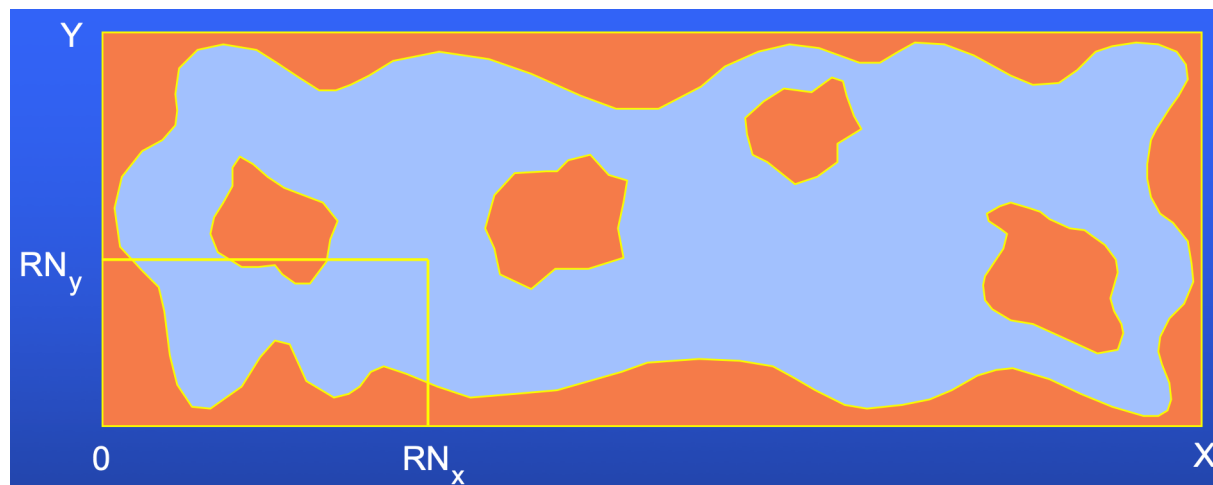
Monte Carlo M&S

- based on (pseudo-)casual number generation
- example: given a high-resolution computer image of a map of an irregularly shaped lake with several islands, determine the water surface area. Assume that the x-y coordinates of every point on the map can be measured.



Monte Carlo M&S

- *Step 1*: enclose the area of interest in the smallest rectangle of known dimensions X and Y
- *Step 2*: generate two uniformly distributed random numbers, RN_x in $[0..X]$ e RN_y in $[0..Y]$
- *Step 3*: If the point of intersection (RN_x, RN_y) falls on the water surface area, add 1 to S (being S the number of hits on the water surface, with $S=0$ at startup)
- *Step 4*: repeat Step 2 and Step 3 for a number N times
- *Step 5*: the estimate of the water surface area is $X \cdot Y \cdot \frac{S}{N}$



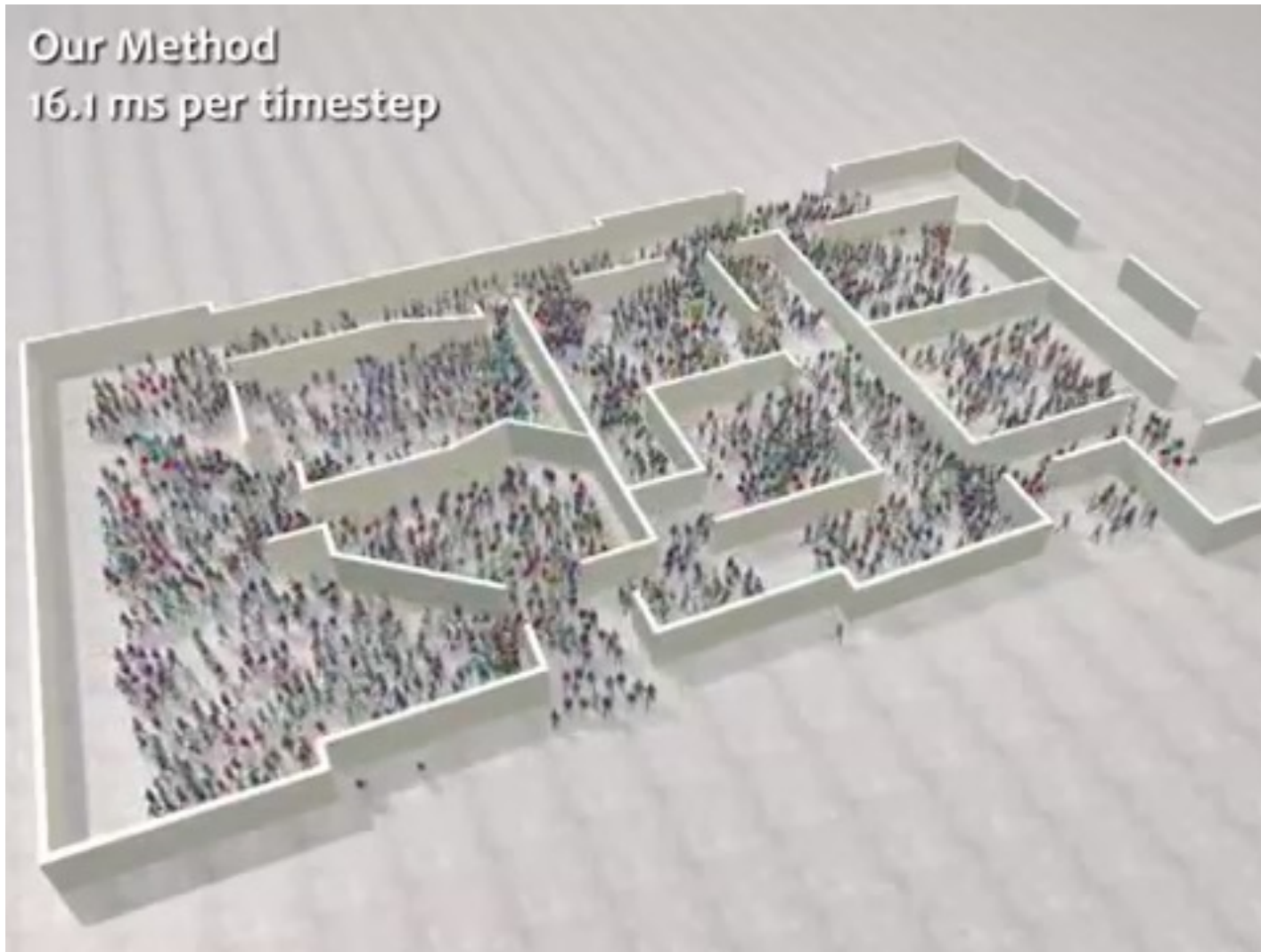
Agent-based M&S

- Agent-based M&S is the one which uses a model representing multiple autonomous entities and their interactions
 - actions are not predefined
 - *agents* representing entities react to the events occurring in their environment
- An agent is:
 - autonomous
 - adaptive
 - reactive
 - proactive
 - “social”

and has the ability to learn (change its behavior based on experience)

Agent-based M&S example

building evacuation



Agent-based M&S example

crowd simulation



Agent-based M&S example

virus spread simulation

powered by NetLogo

Virus

File: New

Export: NetLogo HTML

Mode: Interactive Commands and Code: Bottom

model speed

ticks: 0

number-people 150

setup

go

infectiousness 65 %

chance-recover 75 %

duration 20 weeks

turtle-shape

person

%infected

6.7

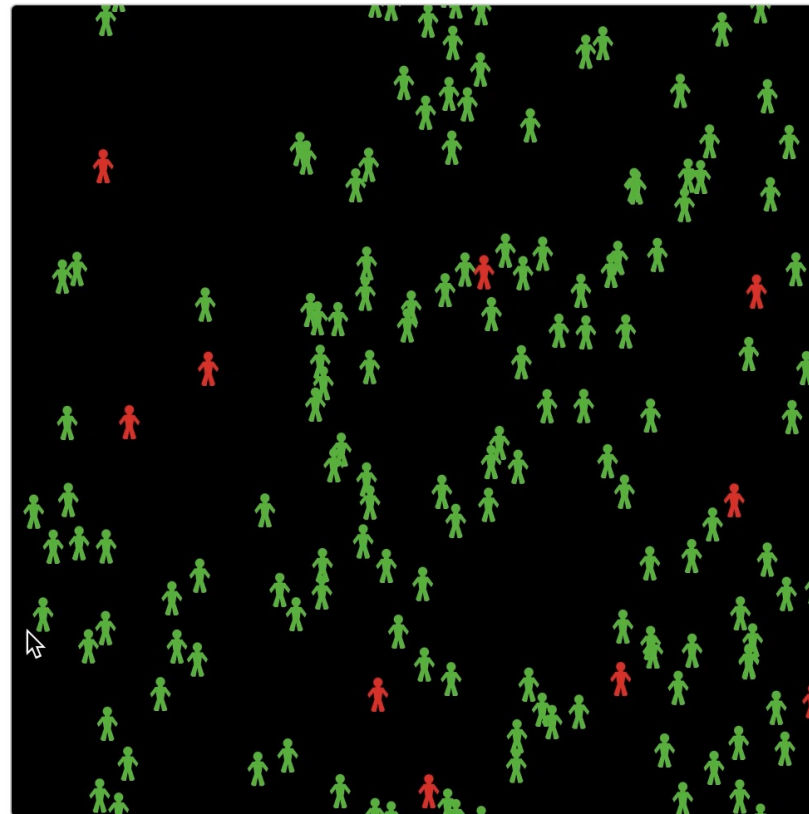
%immune

0

years

0

Populations



Some (not all) application areas



Some (not all) application areas

- Designing and analyzing manufacturing systems
- Evaluating military weapons systems or their logistics requirements
- Determining hardware requirements or protocols for communications networks
- Determining hardware and software requirements for a computer system
- Designing and operating transportation systems such as airports, freeways, ports and subways
- Evaluating designs for service organizations such as call centers, fast-food restaurants, hospitals and post offices
- Analysis and reengineering of business processes
- Determining ordering policies for an inventory system
- Analyzing financial or economic systems